AGRIVIVE / AHAR-CELL

BIHAR / RUNOFF CAPTURE / SERVICEABLE SEDIMENT INTERCEPTOR

SERIES 04 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



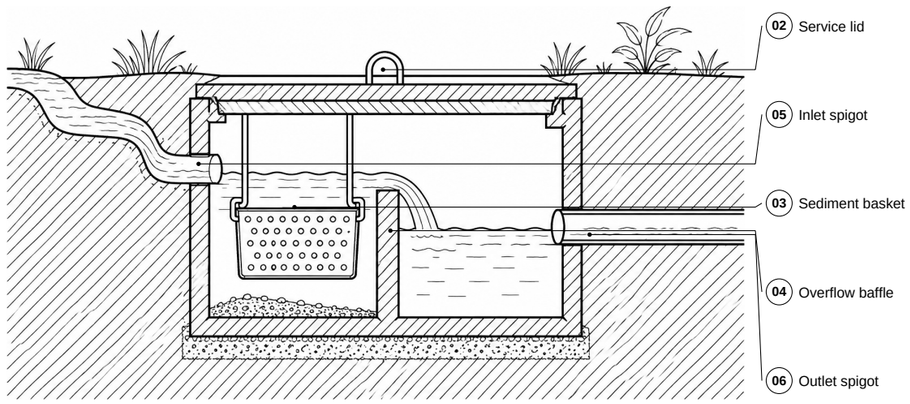
STATE CONTEXT / INDIGENOUS INSPIRATION

Ahar-Pyne networks in south Bihar link runoff diversion channels with storage areas. AHAR-CELL adapts the sequence of interception, settling and transfer into a small serviceable module.

Practice source: Bihar Minor Water Resources Department - Ahar-Pyne systems | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 160 L x 100 W x 120 H

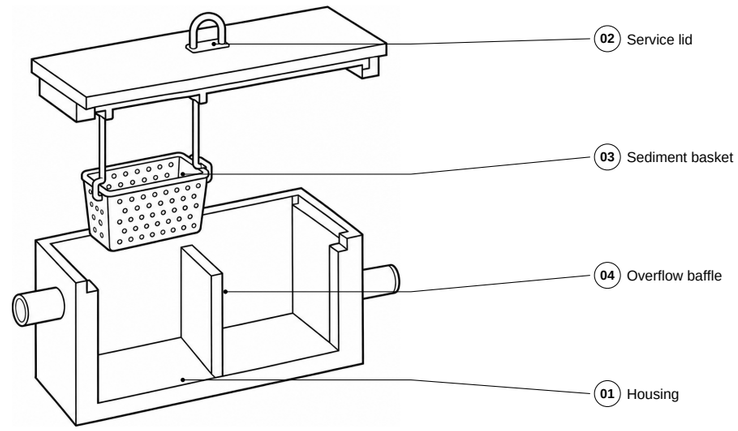


DESIGN / OPERATION NOTES

Runoff enters the left chamber and passes the basket.
Sediment can collect below the basket. Water overtops the baffle to the outlet chamber.
Lift the lid and remove the basket for cleaning; retain accessible service clearance.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
160 L x 100 W x 120 H

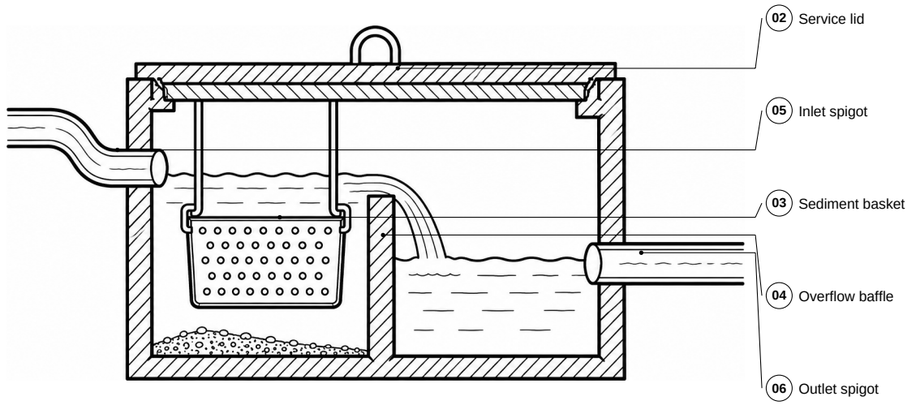


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

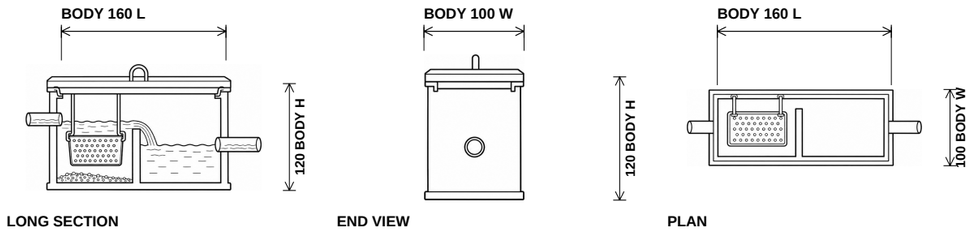
SECTION BASIS / mm
Clear chamber 144 L x 84 W x 100 H before baffle/basket



DESIGN / OPERATION NOTES

Sediment can collect below the basket. Water overtops the baffle to the outlet chamber.
Lift the lid and remove the basket for cleaning; retain accessible service clearance.

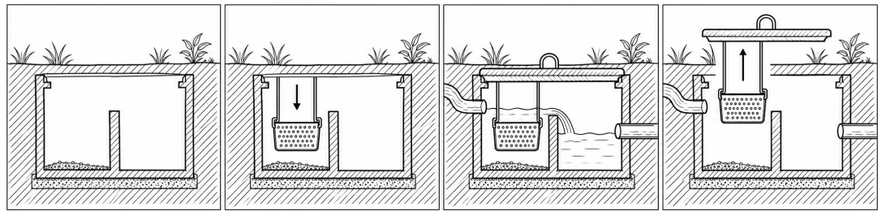
03 ORTHOGRAPHIC STUDIES / mm



LONG SECTION **END VIEW** **PLAN**

Section: 160 L x 120 H. Plan: 160 L x 100 W. Ports and baffle detail TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 SEAT BODY **2 INSERT BASKET** **3 CONNECT / CLOSE** **4 CLEAN BASKET**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External L x W x H	**160 x 100 x 120**
Clear envelope	**144 x 84 x 100**
Port / mesh detail	**TBD**

02 GEOMETRIC CHECK

GROSS CHAMBER ENVELOPE

1,209.6 mL

144 x 84 x 100 / 1000
Subtract baffle, basket, sediment and operating freeboard

03 PARTS / QTY

01 Housing	**1**
02 Service lid	**1**
03 Sediment basket	**1**
04 Overflow baffle	**Integral**
05 Inlet spigot	**Integral**
06 Outlet spigot	**Integral**

Housing/lid/basket: candidate polymer. Screen opening and water-contact suitability TBD.

04 INTERFACES / CHECKS

Check clogging and short-circuit flow around basket.

Confirm inlet / outlet levels and operating freeboard.

Measure actual retained volume after internal features.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-04 / P01	AHAR-CELL	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	04 / 28